LeafLens

AI-Powered Leaf Disease Prediction System

User Manual

Prepared by
Abhijeet Kumar
DA24B001

Contents

1 Welcome to LeafLens **2**
1.1 What Can LeafLens Do? . . . 2

2 Before You Start **2**
2.1 Starting the Application . . . 2
2.2 Application Addresses . . . 2

3 Using LeafLens **3**
3.1 Step 1: Open the Application . . . 3
3.2 Step 2: Select a Leaf Image . . . 3
3.3 Step 3: Submit the Image . . . 3
3.4 Step 4: View the Prediction . . . 3

4 How a Prediction Works **3**

5 Understanding the Prediction **4**
5.1 A Few Practical Suggestions . . . 4

6 Monitoring Dashboard **4**
6.1 What Does the Dashboard Show? . . . 4

7 Troubleshooting **5**
7.1 Viewing Application Logs . . . 5

8 Stopping the Application **5**

9 Important Notes **5**

10 Conclusion **6**

1 Welcome to LeafLens

LeafLens is an AI-powered application that helps users identify possible plant leaf diseases from images.

You do not need any machine learning knowledge to use it. Simply open the application, upload a photograph of a leaf, and wait for the prediction.

The application analyses the image using a trained deep learning model and displays the predicted category along with its confidence score.

This manual explains how to start the application, upload an image, understand the result, and handle common issues.

1.1 What Can LeafLens Do?

With LeafLens, you can:

- Upload an image of a plant leaf.
- Get a predicted disease or healthy-leaf category.
- View the model's prediction confidence.
- Use a simple web-based interface.

LeafLens currently supports 38 categories from the PlantVillage dataset.

Please remember: LeafLens provides an AI-generated prediction. It is intended to support initial identification, not replace advice from an agricultural expert.

2 Before You Start

The application is packaged using Docker Compose. Before starting it, make sure that:

- Docker Desktop is installed.
- Docker Desktop is running.
- The LeafLens project folder is available.
- The required backend and frontend Docker images have been built.

The project uses four services: the frontend, backend, Prometheus, and Grafana.

2.1 Starting the Application

Open a terminal and navigate to the LeafLens project directory.

Build the backend image:

```
docker build -f backend/Dockerfile \
  -t leaflens-backend:latest .
```

Build the frontend image:

```
docker build -f frontend/Dockerfile \
  -t leaflens-frontend:latest .
```

Once both images are built, start the application:

```
docker compose up -d
```

Docker Compose starts the application services in the background.

To check whether the services are running, execute:

```
docker compose ps
```

The backend should report a healthy status before the frontend becomes available.

2.2 Application Addresses

Once the services are running, open the following addresses in your web browser.

Table 1: LeafLens application URLs

Service	Address	Purpose
Frontend	http://localhost:8501	Main LeafLens application.
Backend	http://localhost:8001	Backend API service.
Prometheus	http://localhost:9092	Metrics collection interface.
Grafana	http://localhost:3002	Monitoring dashboard.

For normal use, you only need the frontend address.

3 Using LeafLens

Using the application is straightforward. The main task is to upload a clear image of the leaf you want to examine.

3.1 Step 1: Open the Application

Open your browser and visit:

http://localhost:8501

The LeafLens interface should appear.

3.2 Step 2: Select a Leaf Image

Choose the image upload option on the application screen and select a leaf photograph from your device.

For better results, try to use an image that:

- Clearly shows the leaf.
- Is reasonably well lit.
- Is not excessively blurry.
- Does not hide the affected region.
- Shows the leaf without excessive obstruction.

The backend accepts image uploads up to 10 MB.

3.3 Step 3: Submit the Image

After selecting the image, use the prediction control provided by the interface.

The image is sent to the LeafLens backend, where it is validated and processed by the trained model.

3.4 Step 4: View the Prediction

Once processing is complete, the application displays the prediction result.

The result identifies the category selected by the model and provides its confidence score.

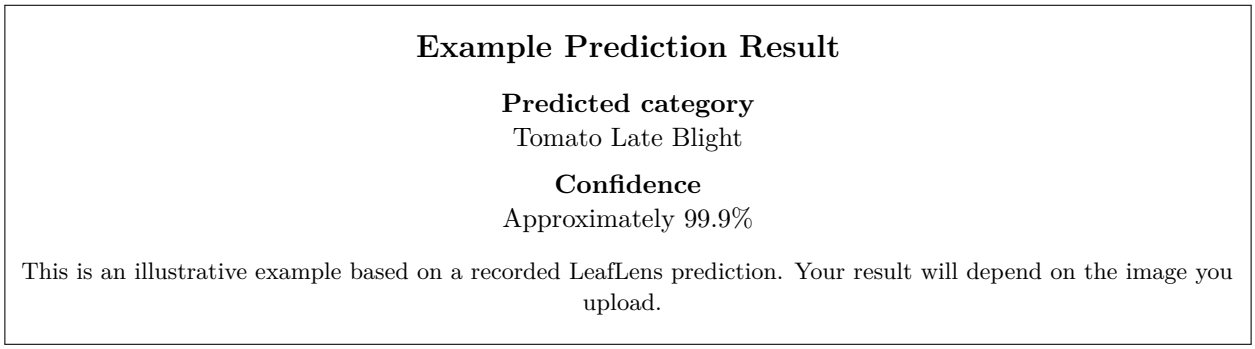


Figure 1: Illustrative prediction result

4 How a Prediction Works

You do not need to understand the internal machine learning process to use LeafLens. However, the following diagram gives a simple idea of what happens after you upload an image.

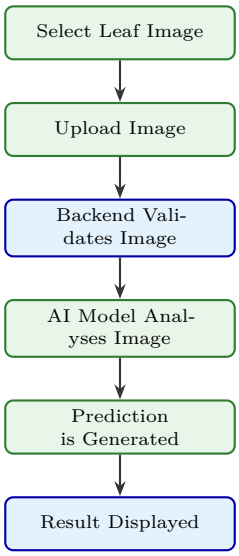


Figure 2: How LeafLens processes an uploaded image

In simple terms, the application checks the uploaded file, prepares the image for the model, runs the classification process, and returns the result.

5 Understanding the Prediction

The predicted category is the class that receives the highest probability from the model.

The confidence score indicates how strongly the model favours that predicted category relative to the other classes it considers.

A high confidence score does not guarantee that the prediction is correct. The model can still make mistakes, especially when an image differs from the images used during training.

For this reason, results should be interpreted carefully.

5.1 A Few Practical Suggestions

- Capture the leaf clearly and avoid motion blur.
- Use adequate lighting.
- Try another photograph if the result seems inconsistent with what you observe.
- Consult an agricultural expert before taking important crop-treatment decisions.

6 Monitoring Dashboard

LeafLens also includes a monitoring dashboard. This part is mainly intended for the project administrator or evaluator rather than an ordinary application user.

Open Grafana at:

`http://localhost:3002`

For the current local demonstration, the default login credentials are:

Field	Value
Username	admin
Password	leaflens_local_admin

The password is configured for local demonstration use. It should be changed before exposing the dashboard outside a trusted local environment.

6.1 What Does the Dashboard Show?

The LeafLens dashboard includes:

- Backend availability.
- Number of incoming API requests.

- Total successful predictions.
- Average prediction confidence.
- P95 API latency.
- HTTP errors.
- Prediction failures.
- Predictions grouped by disease class.

This dashboard helps the administrator understand whether the application is operating normally.

7 Troubleshooting

If the application does not behave as expected, the following table covers some common issues.

Table 2: Common problems and possible solutions

Problem	What to do
Frontend does not open	Check whether Docker Desktop is running and whether the frontend container is active.
Backend is unavailable	Check the backend health status using `http://localhost:8001/health`.
Image upload fails	Make sure the file is a valid image and is not larger than 10 MB.
Prediction fails	Check whether the backend is healthy and inspect its container logs.
Grafana does not load	Check whether the Grafana container is running and whether port 3002 is available.
Dashboard has no data	Verify that Prometheus is running and that the LeafLens backend target is UP.

7.1 Viewing Application Logs

If an issue continues, the project administrator can inspect the Docker logs.

To view all service logs:

```
docker compose logs
```

To inspect only the backend:

```
docker compose logs backend
```

To inspect only the frontend:

```
docker compose logs frontend
```

These commands can help identify startup issues and application errors.

8 Stopping the Application

When you have finished using LeafLens, open the terminal in the project directory and execute:

```
docker compose down
```

This stops and removes the running containers and the Compose network.

The persistent Prometheus and Grafana volumes are not removed by this command, so their stored data is retained.

To stop the services and also remove their persistent volumes, a different command is required:

```
docker compose down -v
```

Use the second command only when you intentionally want to remove the monitoring data as well.

9 Important Notes

A few points are worth keeping in mind while using LeafLens:

- The application currently supports the 38 categories represented in its training dataset.
- The quality of the uploaded image can affect the prediction.
- The confidence score is not a guarantee of correctness.
- The model was evaluated on PlantVillage images, so performance on real-world photographs may differ.
- LeafLens should be used as an initial identification aid, not as a substitute for professional agricultural advice.

10 Conclusion

LeafLens is designed to make plant leaf image classification accessible through a simple interface.

A user only needs to start the application, upload a suitable image, and review the result. The underlying data processing, model inference, and monitoring operations are handled by the application.

We hope this manual makes it easy to get started and understand how to use LeafLens.